

lecture10 Practice Questions and Solutions

Question 1 (q1_sol.py)

Write a program that asks the user to enter their age, and then tells them if they are old enough to vote in the BC provincial election. You must be 18 or older to vote in BC.

Sample run 1

```
Enter your age: 18
You are old enough to vote in the BC provincial election.
```

Sample run 2

```
Enter your age: 15
You are not old enough to vote in the BC provincial election.
```

Sample Solution

```
age = int(input("Enter your age: "))
if age >= 18:
    print("You are old enough to vote in the BC provincial election.")
else:
    print("You are not old enough to vote in the BC provincial election.")
```

Question 2 (q2_sol.py)

Write a program that asks the user to enter the temperature and then tells them what to wear according to these rules:

- If the temperature is 10 degrees or less, they should wear a warm coat.
- If the temperature is greater than 10 degrees and less than 25 degrees, they should wear a light jacket.
- If the temperature is greater than 25 degrees, they should wear a t-shirt.

Sample run 1

```
Enter the temperature: 10
Wear a warm coat.
```

Sample run 2

```
Enter the temperature: 20
Wear a light jacket.
```

Sample run 3

```
Enter the temperature: 31.5
Wear a t-shirt.
```

Sample Solution

```
temp = float(input("Enter the temperature: "))
if temp <= 10:
    print("Wear a warm coat.")
elif 10 < temp < 25:
    print("Wear a light jacket.")
else:
    print("Wear a t-shirt.")
```

Question 3 (q3_sol.py)

The community pool charges \$8 for admission, unless you fall into one of these categories:

- Seniors, aged 65 or older, pay \$5
- Children, aged 6 through 12, pay \$4
- Kids under 6 get in free

Write a program that asks the user their age and then prints their price of admission.

Sample run 1

```
Enter your age: 65
Admission: $5
```

Sample run 2

```
Enter your age: 22
Admission: $8
```

Sample run 3

```
Enter your age: 5
Admission: free!
```

Sample Solution

```
age = int(input("Enter your age: "))
if age >= 65:
    print("Admission: $5")
elif 13 <= age <= 64:
    print("Admission: $8")
elif 6 <= age <= 12:
    print("Admission: $4")
else:
    print("Admission: free!")
```